2c. Content of topics B1 to B6

Topic B1: Cell level systems

B1.1 Cell structures

Summary
Cells are the fundamental units of living organisms. Cells contain many sub-cellular structures that are essential for the functioning of the cell as a whole. Microscopy is used to examine cells and sub-cellular structures.

Underlying knowledge and understanding
Learners should be familiar with cells as the fundamental unit of living organisms, and with the use of light microscopes to view cells. They should also be familiar with some sub-cellular structures, and the similarities and differences between plant and animal cells.

Common misconceptions
Learners commonly have difficulty understanding the concept of a cell as a 3D structure, so this should be addressed during the teaching of this topic.

Tiering
Statements shown in **bold** type will only be tested in the Higher Tier papers. All other statements will be assessed in both Foundation and Higher Tier papers.

Reference	Mathematical learning outcomes	Mathematical skills
BM1.1i	demonstrate an understanding of number, size and scale and the quantitative relationship between units	M2a and M2h
BM1.1ii	use estimations and explain when they should be used	M1d
BM1.1iii	**calculate with numbers written in standard form**	**M1b**

Topic content		Opportunities to cover:		Practical suggestions
Learning outcomes	To include	Maths	Working scientifically	
B1.1a describe how light microscopes and staining can be used to view cells	lenses, stage, lamp, use of slides and cover slips, and the use of stains to view colourless specimens or to highlight different structures/ tissues and calculation of magnification	**M1b**, M1d, M2a, M2h	WS1.2c, WS1.4c, WS1.4d, WS1.4e, WS2a, WS2b, WS2c, WS2d	Investigation of a range of cells using pictures, light micrographs and diagrams. Measure the size and magnification of the cells. (PAG B1, PAG B7) Preparation of cheek cell slides. (PAG B7) Preparation of onion epidermis cells slides. (PAG B7) Use of light microscopes to view plant and animal cells. (PAG B7)
B1.1b explain how the main sub-cellular structures of eukaryotic cells (plants and animals) and prokaryotic cells are related to their functions	nucleus, genetic material, chromosomes, plasmids, mitochondria (contain enzymes for cellular respiration), chloroplasts (contain chlorophyll) and cell membranes (contain receptor molecules, provides a selective barrier to molecules), ribosomes (site of protein synthesis)		WS1.4a, WS2a, WS2b, WS2c, WS2d	Production of 3D model plant and animal cells to illustrate their differences. Investigation of cytoplasmic streaming in Elodea spp. (PAG B6, PAG B7)
B1.1c explain how electron microscopy has increased our understanding of sub-cellular structures	increased resolution in a transmission electron microscope	**M1b**	WS1.1a, WS1.4c, WS1.4d	Comparison of a range of cells using pictures from light and electron micrographs. Comparison of the visible structures visible on light and electron micrographs.